60-question answer key and explanation guide

REVIEW DRAFT. Source-authored explanations retained; independent academic validation is pending.

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1. A patient has a sudden rise in arteriolar sympathetic firing with minimal change in cardiac output. Which immediate variable rises most directly?

Answer: C. Total peripheral resistance.

Explanation: Arteriolar constriction increases resistance and therefore tends to raise arterial pressure even at stable output.

Not A: SA nodal rate concerns cardiac autonomic effects rather than arteriolar action.

Not B: Metabolite clearance depends on local blood flow, not this direct action.

Not D: Sympathetic venoconstriction generally reduces venous compliance.

Source: circulation-2:Q27 | 3, 13, 19, 21

2. The R-R separation is four large squares on a regular standard-speed ECG. What is the heart rate?

Answer: A. 75 beats/min.

Explanation: At 25 mm/s, a large square equals 0.20 s. Four large squares represent 0.8 s per beat, giving 60/0.8 = 75 beats/min.

Not B: 120 beats/min would require 2.5 large squares.

Not C: 60 beats/min would require five large squares between R waves.

Not D: 90 beats/min would require about 3.33 large squares.

Source: lecture-7:Q42 | 13-14

3. An ECG develops a new shift of the ST segment above or below the baseline. Which process specifically makes this segment clinically important?

Answer: A. Myocardial ischaemia or inadequate oxygenation can alter ST level.

Explanation: Ventricular ischaemia or hypoxia can disturb membrane potentials and cause ST elevation or depression, so the segment is useful in recognising myocardial injury patterns.

Not B: Atrial depolarisation is the P wave and normally precedes ST.

Not C: Preload is a filling variable and does not directly set ST height.

Not D: ST displacement is an electrical finding, not the normal marker of valve closure.

Source: lecture-7:Q27 | 9

4. Portal hypertension promotes ascites partly by increasing which force?

Answer: B. Hydrostatic pressure in hepatic sinusoids.

Explanation: Higher sinusoidal hydrostatic pressure increases fluid filtration and hepatic lymph formation; excess can enter the peritoneal cavity.

Not A: Higher plasma oncotic pressure would oppose outward fluid filtration.

Not C: Improved lymphatic clearance would tend to oppose accumulation.

Not D: Oxygen tension is not the main Starling-force mechanism.

Source: lecture-6:Q44 | 37

5. Which contrast best distinguishes forward failure from backward failure?

Answer: A. Forward reduces delivery; backward causes venous congestion.

Explanation: Forward failure describes inadequate delivery to downstream tissues, whereas backward failure describes elevated pressure and congestion upstream of a failing chamber.

Not B: The two effects are reversed in this statement.

Not C: Reduced flow and congestion, not excessive output, are the defining patterns.

Not D: Either side can produce both forward and backward consequences.

Source: circulation-3:Q48 | Circulation 3, slides 32-33

6. After severe head injury, a patient has high blood pressure, a slow pulse and irregular breathing. What does this combination suggest?

Answer: B. A dangerous Cushing response to intracranial compression.

Explanation: Hypertension, reflex bradycardia and abnormal respiration form the Cushing triad, a late sign of severe intracranial pressure.

Not A: Exercise more commonly raises both rate and output.

Not C: Local muscle dilation does not explain the full neurological triad.

Not D: Atrial overfilling usually causes tachycardia in the lecture's account.

Source: circulation-2:Q56 | 32-36

7. What directly promotes pulmonary oedema in left-sided backward heart failure?

Answer: C. Raised pulmonary venous and capillary pressure.

Explanation: Left ventricular failure increases left atrial and pulmonary venous pressures, driving filtration into the lungs.

Not A: Lower capillary hydrostatic pressure would oppose filtration.

Not B: Systemic venous pressure is more closely linked to right-sided failure.

Not D: Right atrial pressure alone does not explain pulmonary oedema.

Source: circulation-3:Q52 | Circulation 3, slide 33; Chapter 21 ?3

8. Why does the fetal cardiovascular system divert much blood away from the pulmonary circuit before birth?

Answer: D. The placenta oxygenates blood; fetal lungs resist flow.

Explanation: The placenta performs gas exchange before birth; fluid-filled fetal lungs have high resistance, favouring right-to-left shunts.

Not A: The fetal left ventricle contributes significantly to systemic output.

Not B: Pulmonary resistance is high before aeration.

Not C: Umbilical arteries carry relatively deoxygenated blood toward placenta.

Source: lecture-6:Q53 | 42-44; Chapter 27 ?1

9. After bleeding stops, both haemoglobin and plasma-protein concentrations fall as circulating volume recovers. What explains this?

Answer: A. Added fluid dilutes the remaining blood components.

Explanation: Fluid recruited from tissues or given intravenously restores volume before red-cell and protein mass recover, producing dilution.

Not B: New red-cell production raises rather than lowers haemoglobin mass.

Not C: Renal filtration is generally depressed during acute severe shock.

Not D: Venoconstriction redistributes blood rather than removing haemoglobin.

Source: circulation-3:Q35 | Circulation 3, slides 24

10. An ECG electrode detects a voltage on the skin even though the impulse originates in the myocardium. What makes this possible?

Answer: C. Cardiac electrical currents spread through conducting body tissues.

Explanation: Depolarisation and repolarisation create voltage differences that propagate through the conductive tissues of the body and can be detected at the skin.

Not A: Mechanical pulsation is distinct from electrical potential recorded by ECG electrodes.

Not B: Surface ECG electrodes do not directly measure intravascular blood pressure.

Not D: The electrodes sit on intact skin, not within the myocardium.

Source: lecture-7:Q1 | 2

11. Heart rate rises from 70 to 80 beats/min while stroke volume stays at 70 mL. By how much does cardiac output increase?

Answer: B. By 0.70 litres per minute.

Explanation: The increase is 10 beats/min x 70 mL/beat = 700 mL/min, or 0.70 L/min.

Not A: 0.80 L/min treats the extra beats as a larger volume.

Not C: 0.49 L/min does not follow from the given rate change.

Not D: 1.40 L/min doubles the actual change.

Source: circulation-2:Q20 | 1, 3

12. At unchanged blood density, vessel diameter and viscosity, what happens to Reynolds number if average blood velocity doubles?

Answer: A. A twofold increase.

Explanation: Reynolds number is proportional to blood velocity when density, viscosity and vessel diameter remain fixed. Doubling velocity therefore doubles Reynolds number and makes turbulence more likely.

Not B: A halving would require the velocity-dependent term to decrease.

Not C: Velocity enters the Reynolds relation linearly, not as its square.

Not D: Reynolds number changes when velocity changes at fixed other variables.

Source: circulation-1:Q54 | 65-66

13. Most sympathetic constrictor fibres release noradrenaline onto arteriolar smooth muscle. Which receptor mainly mediates constriction?

Answer: D. The alpha-1 adrenergic receptor.

Explanation: Noradrenaline binds alpha1 receptors on arteriolar smooth muscle, raising intracellular Ca^2+ and contracting the muscle. This raises resistance and arterial pressure.

Not A: beta2 activation typically dilates muscle arterioles.

Not B: M2 receptors mediate vagal slowing of the heart.

Not C: H2 receptors respond to histamine, which dilates arterioles.

Source: circulation-2:Q25 | 13-14, 19

14. A harsh systolic murmur develops during blood ejection through a narrowed aortic valve. What mechanism explains it?

Answer: B. Turbulent forward flow through a stenotic outflow orifice.

Explanation: Aortic stenosis accelerates blood through a narrow outlet during systole, causing a turbulent ejection murmur.

Not A: The aortic valve opens when ventricular pressure exceeds aortic pressure in systole.

Not C: Mitral regurgitation usually occurs during systole, not diastole.

Not D: Laminar flow through normal valves ordinarily produces no such murmur.

Source: circulation-1:Q32 | 31-33; Chapter 5 ?7

15. What distinguishes the normal x descent from the y descent in a JVP trace?

Answer: A. x: atrial relaxation; y: emptying through the open tricuspid.

Explanation: The x descent reflects atrial relaxation and downward tricuspid movement; the y descent follows tricuspid opening as blood empties rapidly.

Not B: Atrial contraction creates the a wave.

Not C: Both descents reflect right atrial pressure changes rather than the proposed events.

Not D: Left-heart valve events do not define these JVP descents.

Source: circulation-1:Q70 | 24-25; Chapter 11 ?4

16. At 25 mm/s, a PR interval extends across 4.5 small squares. Which interpretation is most accurate?

Answer: D. 0.18 s, inside the stated PR-interval range.

Explanation: The horizontal duration is 4.5 x 0.04 = 0.18 s. This lies within the 0.12-0.20 s interval given in the lecture.

Not A: This is not the product of 4.5 and 0.04 s.

Not B: This incorrectly halves the duration obtained from 4.5 small squares.

Not C: This incorrectly assigns 0.20 s to each small square.

Source: lecture-7:Q37 | 7, 13

17. Why does fetal haemoglobin load oxygen effectively at the placental interface?

Answer: B. Its weaker 2,3-BPG binding increases oxygen affinity.

Explanation: HbF interacts less with 2,3-BPG, shifting its oxygen dissociation curve left and favouring oxygen uptake from maternal blood.

Not A: Stronger 2,3-BPG binding would lower oxygen affinity.

Not C: Fetal haemoglobin still contains iron-bearing haem groups.

Not D: Oxygen binding is reversible, not permanent.

Source: lecture-6:Q58 | 45

18. A patient develops headache that is worse upright after lumbar CSF sampling. Which mechanism is most likely?

Answer: A. Persistent CSF leakage lowers intracranial CSF pressure.

Explanation: Leakage from the dural puncture can reduce CSF pressure and cause a characteristic postural headache.

Not B: BBB disruption is not the usual cause of post-lumbar-puncture headache.

Not C: Production does not normally surge to cause this symptom.

Not D: The foramina are far from the lumbar puncture site.

Source: lecture-6:Q11 | 17; Chapter 22 ?5

19. During severe hepatic hypoperfusion, cells near the central vein are especially vulnerable to hypoxia. Why?

Answer: C. They are farthest downstream from the portal triads.

Explanation: Centrilobular cells receive blood after it has travelled from portal triads, making them less oxygenated and vulnerable.

Not A: Centrilobular hepatocytes remain within the sinusoidal perfusion system.

Not B: Oxygenated hepatic arterial inflow enters near the portal triads.

Not D: Blood oxygen content generally falls as it traverses sinusoids.

Source: lecture-6:Q41 | 35

20. Following a substantial meal, intestinal blood flow rises for several hours. What primarily drives this response?

Answer: D. Local metabolic activity and absorptive demand.

Explanation: Postprandial metabolic and absorptive activity promotes local gastrointestinal hyperaemia; the lecture notes it can last about three hours.

Not A: Sympathetic constriction usually opposes the flow increase.

Not B: Absorption increases rather than abolishes local work.

Not C: Portal flow continues to deliver absorbed substances to liver.

Source: lecture-6:Q42 | 36

21. Two identical limbs have arterial flow interrupted for one minute and five minutes respectively. Which reperfusion pattern is expected?

Answer: B. The five-minute block produces a larger, longer surge.

Explanation: Longer interruption creates a greater oxygen debt and metabolite buildup, producing a larger and more prolonged reactive response.

Not A: Reactive hyperaemia is expected when flow is restored.

Not C: The amount of accumulated metabolites depends on occlusion duration.

Not D: A shorter block generally produces a smaller oxygen debt.

Source: circulation-2:Q14 | 11

22. Two Frank-Starling curves are plotted against identical end-diastolic volumes. What does an upward shift of the entire curve usually indicate?

Answer: A. Greater contractility at the same preload.

Explanation: Positive inotropy raises stroke volume or output at the same filling level, shifting the Frank-Starling function upward.

Not B: Higher afterload tends to impair ejection.

Not C: Valve closure timing is not the definition of an upward Starling curve.

Not D: Lower force would shift the curve downward, not upward.

Source: circulation-1:Q37 | 39-42

23. During a stress response, circulating adrenaline dilates vessels in working skeletal muscle while constricting many skin vessels. Why?

Answer: B. Different vascular beds express different adrenergic receptors.

Explanation: beta2 effects can favour dilation in skeletal muscle, while alpha1 effects cause constriction in many skin and gut vessels.

Not A: Muscle vessels respond because they possess adrenergic receptors.

Not C: The difference is due mainly to receptor distribution rather than selective circulating delivery.

Not D: Plasma volume differences do not explain opposite receptor responses.

Source: circulation-2:Q32 | 12-13, 30-32

24. An older patient faints shortly after standing to urinate at night. Which phenomenon does this episode most closely illustrate?

Answer: B. Micturition syncope linked to reflex hypotension.

Explanation: Micturition syncope is fainting associated with urination, often after rising at night, due to a reflex pressure fall.

Not A: Carotid sinus syncope requires stimulation of a sensitive carotid region.

Not C: There is no evidence of infection-related shock in the stem.

Not D: Normal standing to urinate is not high positive acceleration.

Source: circulation-3:Q13 | Chapter 19 ?4 (supplementary)

25. In severe haemorrhagic shock, why does sympathetic constriction of splanchnic vessels help initially?

Answer: D. It redistributes flow and mobilises stored venous blood.

Explanation: The splanchnic bed is a large blood reservoir; vasoconstriction redirects perfusion and recruits stored blood toward central circulation.

Not A: Mucosal flow may decrease, risking ischaemia.

Not B: Extraction is not necessarily complete.

Not C: Severe or prolonged constriction can contribute to intestinal injury.

Source: lecture-6:Q43 | 36

26. An examination reveals raised JVP, a tender enlarged liver, ascites and ankle oedema. Which haemodynamic change unifies these?

Answer: A. High systemic venous pressure behind the right ventricle.

Explanation: Right ventricular backward failure elevates right atrial and systemic venous pressures, causing jugular distension, hepatic congestion and dependent oedema.

Not B: Higher systemic venous pressure promotes, rather than reduces, filtration.

Not C: Isolated pulmonary venous congestion principally reflects left-sided failure.

Not D: Right-sided backward failure increases rather than decreases venous pressure.

Source: circulation-3:Q56 | Circulation 3, slides 33

27. An isolated SA node fires at about 100 beats/min, whereas the same person's resting pulse was about 70. Which interpretation best fits?

Answer: C. Resting vagal activity had restrained SA nodal firing.

Explanation: With no nerve input the SA node fires at about 100 beats/min. At rest the vagus releases acetylcholine continuously (vagal tone), holding the rate down to about 60-80. Blocking the vagus, for example with atropine, lets the rate rise.

Not A: Atrial conduction is not the main cause of the slower intrinsic SA pacemaker rate.

Not B: The isolated rate shows the pacemaker is capable of firing faster.

Not D: Sympathetic activity usually increases SA nodal firing.

Source: circulation-2:Q30 | 20

28. Immediately after the first effective breaths, what happens to pulmonary vascular resistance?

Answer: D. It falls as the lungs expand and oxygen tension rises.

Explanation: Lung aeration and increased alveolar oxygen rapidly lower pulmonary vascular resistance, increasing pulmonary blood flow.

Not A: Alveolar oxygen generally relieves pulmonary hypoxic vasoconstriction.

Not B: Pulmonary adaptation begins with the first breaths.

Not C: Pulmonary and systemic circuits retain very different resistance values.

Source: lecture-6:Q59 | 45; Chapter 27 ?4

29. A technician places V1 at the fourth intercostal space. Where should the electrode be positioned horizontally?

Answer: C. At the right sternal border.

Explanation: V1 is positioned in the fourth intercostal space at the right edge of the sternum; V2 lies at the left sternal edge at the same level.

Not A: The left midclavicular line in the fifth space is the usual site of V4.

Not B: The anterior axillary position is used for V5.

Not D: The midaxillary position is used for V6.

Source: lecture-7:Q11 | 5; Chapter 6 ?3

30. Two beats have different QRS heights. A student concludes the taller complex produced a larger stroke volume. What is the main flaw?

Answer: D. QRS height reflects voltage, not the amount of blood ejected.

Explanation: The ECG measures electrical potentials. A larger recorded deflection may reflect electrical orientation or amplitude; it does not directly quantify mechanical stroke volume.

Not A: Duration is measured horizontally on ECG paper, not from complex height.

Not B: A surface ECG records electrical potential rather than ventricular pressure.

Not C: Venous return is a haemodynamic variable not directly measured by the tracing.

Source: lecture-7:Q2 | 2, 6

31. Which finding would most strongly suggest pulmonary rather than isolated dependent leg oedema?

Answer: D. Lung crackles and dyspnoea with left-heart failure.

Explanation: Left ventricular failure raises pulmonary venous pressure, causing lung interstitial and alveolar fluid with breathlessness and crackles.

Not A: Dependent leg oedema is more commonly systemic venous or local hydrostatic.

Not B: Pitting alone does not localise oedema to the lungs.

Not C: Postmastectomy swelling suggests lymphatic obstruction.

Source: circulation-1:Q91 | 90-95; Chapter 14 ?9

32. Why can longstanding filarial lymphoedema become firm and poorly pitting?

Answer: A. Chronic protein-rich interstitial fluid promotes tissue fibrosis.

Explanation: Persistent failure of lymphatic drainage leaves protein-rich fluid in the interstitium, promoting chronic inflammation and fibrosis. This explains why longstanding lymphoedema becomes firm and often non-pitting.

Not B: Rapid reabsorption would tend to reduce swelling and protein accumulation.

Not C: Chronic interstitial protein deposition can drive fibrosis despite any venous tone changes.

Not D: The chronic fibrosis is linked to sustained lymphatic obstruction, not a brief arterial-pressure change.

Source: circulation-1:Q97 | 95; Chapter 15 ?7

33. Immediately after rapid loss of a large volume of whole blood, haemoglobin concentration is unexpectedly unchanged. Why?

Answer: C. Red cells and plasma were lost together.

Explanation: Acute whole-blood loss removes red cells and plasma proportionately, so concentration may remain unchanged before fluid replacement.

Not A: Red-cell mass falls with whole-blood loss.

Not B: Early dilution has not yet occurred in this scenario.

Not D: Erythropoiesis takes days to weeks, not minutes.

Source: circulation-3:Q34 | Circulation 3, slides 24

34. A patient with left-sided forward heart failure reports fatigue and produces very little urine. Which mechanism links the two?

Answer: B. Reduced systemic perfusion limits muscle and renal blood flow.

Explanation: Reduced blood ejected from the left heart impairs delivery to skeletal muscle and kidneys, causing fatigue and oliguria.

Not A: Higher venous pressure does not directly improve GFR.

Not C: The primary forward mechanism is reduced output, not pulmonary congestion.

Not D: The syndrome involves reduced rather than excessive systemic delivery.

Source: circulation-3:Q53 | Circulation 3, slides 32

35. A recording shows that cardiac vagal stimulation lowers the firing rate of the sinoatrial node. Which additional effect is expected?

Answer: C. Slower impulse conduction through the AV node.

Explanation: Parasympathetic acetylcholine acts at muscarinic M2 receptors to slow both SA nodal firing and AV nodal conduction.

Not A: beta1 activation describes sympathetic, not vagal, stimulation.

Not B: AV conduction is slowed rather than accelerated by vagal action.

Not D: The major cardiac pacemaker effect is not mediated by alpha1.

Source: circulation-2:Q29 | 20

36. A patient has hydrocephalus due to impaired absorption by arachnoid villi without an obstructed ventricular pathway. What type is this?

Answer: C. Communicating hydrocephalus.

Explanation: Communicating hydrocephalus occurs when CSF reaches subarachnoid space but reabsorption is impaired. The ventricular pathway remains open, so the defect is in absorption rather than transit.

Not A: Noncommunicating hydrocephalus results from blocked ventricular drainage.

Not B: Aqueductal stenosis is a specific obstruction.

Not D: Haemorrhage may contribute to hydrocephalus but is not the classification described.

Source: lecture-6:Q10 | 16

37. What conservation principle underlies Fick measurement of cardiac output?

Answer: C. Oxygen use equals flow x arterial-venous oxygen difference.

Explanation: By mass balance, oxygen uptake across lungs and consumption by tissues equal cardiac output multiplied by arterial-venous oxygen content difference.

Not A: Tissues extract oxygen, so venous content is generally lower.

Not B: Extraction and flow together determine total oxygen consumption.

Not D: The body consumes oxygen but does not produce it.

Source: circulation-1:Q45 | 52-53

38. What keeps tissue fluid moving through collecting lymph vessels toward the great veins?

Answer: A. Valves, muscle movement and lymphatic contractions.

Explanation: One-way valves, skeletal-muscle compression, respiratory movement and lymphatic smooth-muscle contraction promote forward flow. The valves prevent reflux between compressions, making the low-pressure pumping effective.

Not B: Whole vessels do not diffuse through tissues.

Not C: Backflow is limited by valves rather than required for transport.

Not D: Lymph flow is not driven directly by systemic arterial pressure.

Source: circulation-1:Q94 | 60; Chapter 15 ?3

39. An arteriole's radius falls to half its previous value; other variables stay constant. How does its resistance change?

Answer: D. A factor of 16.

Explanation: Poiseuille resistance varies with 1/r4; halving radius raises resistance by 24 = 16. The radius effect is disproportionately large because it is raised to the fourth power.

Not A: A doubling assumes linear radius dependence.

Not B: Fourfold assumes a square-law dependence.

Not C: Eightfold assumes a cubic rather than fourth-power relation.

Source: circulation-1:Q55 | 68-70

40. A patient with left-sided heart failure feels breathless when lying flat but improves after sitting up. Why?

Answer: A. Lying flat shifts peripheral blood toward congested lungs.

Explanation: Supine positioning returns peripheral blood to the thorax; a failing left ventricle cannot handle the extra flow, increasing pulmonary venous congestion.

Not B: Central thoracic volume commonly increases rather than decreases supine.

Not C: A higher pulmonary capillary pressure would worsen breathlessness.

Not D: Sitting upright tends to reduce thoracic venous return.

Source: circulation-3:Q55 | Circulation 3, slides 33; Chapter 21 ?3

41. A regular rhythm has 24 small squares between adjacent R waves at standard paper speed. Approximately what is the heart rate?

Answer: C. 63 beats/min.

Explanation: The rate in a regular rhythm is 1500 divided by the R-R separation in small squares: 1500/24 = 62.5 beats/min, approximately 63.

Not A: 50 beats/min corresponds to about 30 small squares between R waves.

Not B: 80 beats/min corresponds to about 18.75 small squares between R waves.

Not D: 60 beats/min corresponds to about 25 small squares between R waves.

Source: lecture-7:Q41 | 13-14

42. What does the normal jugular venous 'v' wave mainly represent?

Answer: D. Atrial filling while tricuspid is closed.

Explanation: Systemic venous return fills the right atrium during ventricular systole while the tricuspid valve remains closed, creating the v wave.

Not A: Pulmonary valve closure is near S2.

Not B: Atrial depolarisation precedes the a wave.

Not C: The v wave is defined by right atrial filling rather than aortic ejection.

Source: circulation-1:Q69 | 24-26; Chapter 11 ?4

43. What is detected by the fingers when palpating the radial pulse?

Answer: B. An arterial wall-distension pressure wave.

Explanation: Each ventricular ejection creates an arterial pressure/distension wave that travels along the arteries. This pressure wave moves faster than the bulk blood itself.

Not A: Venous blood does not directly pulse into systemic arteries.

Not C: Red cells are not separately felt during palpation.

Not D: The palpable pulse is mechanical rather than an electrical conduction wave.

Source: circulation-1:Q66 | 22-23

44. V2 has been placed correctly and V4 is at its usual site. Where should V3 be attached?

Answer: D. Halfway between the V2 and V4 electrodes.

Explanation: V3 is placed between the already positioned V2 and V4, so the anatomical line between those contacts determines its location.

Not A: The midaxillary line at V4 level is used for V6.

Not B: The anterior axillary line at V4 level is used for V5.

Not C: The fourth right parasternal position belongs to V1.

Source: lecture-7:Q12 | 5; Chapter 6 ?3

45. Using Bazett's formula, QTc = QT/?RR, what is QTc when QT is 0.36 s and RR is 0.81 s?

Answer: D. 0.40 s, because QT is divided by 0.90.

Explanation: Since ?0.81 = 0.90, Bazett's correction is 0.36/0.90 = 0.40 s. The correction adjusts the observed QT for heart rate.

Not A: Multiplying by RR is not Bazett's rate-correction formula.

Not B: QT should be divided by ?RR, not multiplied by it.

Not C: Adding QT and RR has no role in the stated equation.

Source: lecture-7:Q32 | 12

46. Where is the largest controllable fall in systemic vascular pressure normally produced?

Answer: C. Across resistance arterioles.

Explanation: Small arteries and arterioles provide substantial resistance, producing a major pressure drop from arterial to capillary levels.

Not A: Large veins carry blood at low pressure and resistance.

Not B: Right atrial pressure is low but does not create the main distributed arterial drop.

Not D: Large elastic arteries conduct flow with relatively small resistance.

Source: circulation-1:Q65 | 57, 61, 76; Chapter 10 ?9

47. Why does the sinoatrial node ordinarily set the heart's rhythm rather than latent pacemakers?

Answer: B. It has the fastest intrinsic depolarisation rate.

Explanation: The SA node normally reaches threshold fastest and overdrive-suppresses slower latent pacemakers. Faster SA-node impulses normally suppress slower escape pacemakers.

Not A: Action potential duration is not the primary reason for the fastest spontaneous rate.

Not C: Coronary perfusion is not determined by pulmonary venous timing.

Not D: Pacemaker dominance does not depend on muscle mass.

Source: circulation-1:Q12 | 7-9; Chapter 3 ?3

48. Which sequence correctly traces normal electrical conduction from atria into ventricular myocardium?

Answer: B. SA node -> AV node -> His bundle -> Purkinje fibres.

Explanation: The SA node initiates impulses that reach the AV node, traverse the His-bundle branches and Purkinje system, then depolarise ventricles.

Not A: Purkinje fibres come after the AV-His pathway.

Not C: The SA node normally initiates atrial excitation.

Not D: Ventricular Purkinje fibres conduct after, not before, atrial activation.

Source: circulation-1:Q11 | 7-10

49. Why can coordinated cardiac pumping continue without a new external nerve impulse for every beat?

Answer: B. Intrinsic pacemaker cells generate rhythmic impulses.

Explanation: Specialised cardiac pacemaker cells depolarise spontaneously; autonomic nerves modulate rather than initiate each heartbeat.

Not A: Atrial conduction is not limited to expiration.

Not C: The SA node produces electrical impulses, not direct ejection pressure.

Not D: Some conducting cells possess spontaneous automaticity.

Source: circulation-1:Q4 | 7-10

50. A patient has near-normal resting cardiac output but very high ventricular filling pressures and pulmonary congestion. Which statement fits best?

Answer: A. Heart failure can occur with normal output.

Explanation: Heart failure includes states where output is maintained only at abnormally high filling pressures, causing congestion.

Not B: High filling pressures can be evidence of dysfunction rather than normality.

Not C: Diastolic failure can occur with preserved ejection fraction.

Not D: A near-normal output can be maintained at harmful filling pressures.

Source: circulation-3:Q47 | Circulation 3, slide 31; Chapter 21 ?1

51. What event ends isovolumetric contraction and begins ventricular ejection?

Answer: D. Ventricular pressure exceeds the outflow artery pressure.

Explanation: Semilunar valves open when ventricular pressure exceeds aortic or pulmonary arterial pressure. The pressure reversal is the immediate mechanical trigger for valve opening.

Not A: This pressure relation opens the AV valves during filling.

Not B: Atrial systole precedes isovolumetric contraction.

Not C: Lower ventricular than aortic pressure holds the aortic valve closed.

Source: circulation-1:Q20 | 14-17

52. Which pressure difference primarily drives blood back to the right atrium from the body?

Answer: C. Systemic filling pressure minus right atrial pressure.

Explanation: Venous return depends on the upstream systemic filling pressure relative to right atrial pressure and the resistance to venous return.

Not A: Aortic-right ventricular systolic pressure is not the venous return driving gradient.

Not B: That difference is a pulse-pressure-related comparison.

Not D: Pulmonary venous pressure does not define systemic venous return.

Source: circulation-1:Q74 | 97; Chapter 12 ?3

53. What can follow when the right ventricle fails to deliver adequate forward flow to the lungs?

Answer: B. Less blood reaches the left heart for systemic output.

Explanation: Reduced right ventricular output limits pulmonary flow and subsequent return to the left atrium, compromising left-sided filling and delivery.

Not A: The right heart cannot deliver excess flow when severely failing.

Not C: Reduced pulmonary transit cannot raise systemic oxygen delivery.

Not D: Less pulmonary venous return tends to decrease left-sided filling.

Source: circulation-3:Q54 | Circulation 3, slides 32

54. Where does the ductus arteriosus carry much fetal right ventricular output?

Answer: B. From pulmonary trunk into descending aorta.

Explanation: High prenatal pulmonary resistance diverts much right ventricular output through ductus arteriosus into the aorta.

Not A: Right-to-left atrial passage occurs at the foramen ovale.

Not C: Left ventricular output does not enter the portal vein.

Not D: Umbilical venous diversion occurs at the ductus venosus.

Source: lecture-6:Q57 | 43-44; Chapter 27 ?2

55. At an otherwise unchanged capillary, how would increased interstitial protein concentration affect fluid movement?

Answer: D. It strengthens the outward oncotic force favouring filtration.

Explanation: More interstitial protein increases interstitial colloid osmotic pressure, tending to pull fluid outward from the capillary and increase filtration.

Not A: Increased proteins outside the capillary attract rather than remove interstitial water.

Not B: Plasma protein concentration is held constant; the changed force is interstitial.

Not C: A change in interstitial protein does not directly lower capillary pressure.

Source: circulation-1:Q83 | 85-88

56. Electrical stimulation of the medullary pressor area increases arterial pressure. What is the principal neural pathway?

Answer: D. Excitation of sympathetic neurons in the spinal cord.

Explanation: The pressor area sends excitatory fibres down to the sympathetic neurons in the spinal cord. More sympathetic output constricts arterioles and veins and speeds and strengthens the heart, so pressure rises.

Not A: Carotid sinus fibres carry sensory information towards the medulla.

Not B: Inhibiting sympathetic neurons would tend to lower pressure.

Not C: Pressor-area activity does not require direct pacemaker inhibition.

Source: circulation-2:Q22 | 16, 19, 21

57. Which pair correctly identifies the lung's respiratory and supporting arterial circulations?

Answer: C. Pulmonary arteries: gas exchange; bronchial arteries: support.

Explanation: Pulmonary arteries carry blood for gas exchange, while bronchial arteries from the aorta nourish lung supporting structures.

Not A: Bronchial arteries primarily nourish lung tissue; coronary arteries nourish heart.

Not B: The functional roles of pulmonary and bronchial supplies are reversed.

Not D: Pulmonary veins return oxygenated blood; aortic veins are not a normal supply.

Source: lecture-6:Q29 | 31

58. A patient has low circulating volume. Which hormone chiefly increases renal water retention and can constrict vessels at high concentration?

Answer: A. Vasopressin, also called ADH.

Explanation: ADH promotes renal water conservation and has a vasoconstrictor action when its level is high.

Not B: ANP promotes sodium and water loss.

Not C: VIP is a vasodilator rather than a water-retaining hormone.

Not D: Prostacyclin dilates vessels and limits platelet aggregation.

Source: circulation-2:Q34 | 12, 29

59. Why might severe anaemia produce a cardiac flow murmur despite structurally normal valves?

Answer: A. Lower viscosity and increased flow favour turbulence.

Explanation: Anaemia lowers viscosity and may increase cardiac output, raising the tendency to turbulent flow and a functional murmur.

Not B: The murmur is not due to forced aortic valve closure.

Not C: Anaemia decreases viscosity rather than increasing it.

Not D: Conduction block is not the principal mechanism of flow murmurs.

Source: circulation-1:Q31 | 31-34

60. During ventricular diastole, what would an increase in peripheral resistance do to the rate of aortic pressure decline if compliance were unchanged?

Answer: C. Slow the decline by reducing downstream runoff.

Explanation: Greater downstream resistance reduces diastolic runoff from the elastic arterial reservoir; at unchanged compliance, aortic pressure falls more slowly.

Not A: Greater resistance normally reduces runoff rather than increasing it.

Not B: Diastolic decay depends in part on resistance and arterial compliance.

Not D: Higher resistance slows forward runoff but does not reverse normal arterial flow.

Source: circulation-1:Q59 | 76-78; Chapter 10 ?2